

Lab 1 – PatientPath

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1 Introduction

The healthcare system continues to struggle with efficient patient information exchange despite advances in digital technology. Approximately 65% of medical record transfers still rely on fax rather than modern digital solutions (Office of the National Coordinator for Health Information Technology [ONC], 2023). Consistent data sharing also remains limited, particularly among long-term care and behavioral health organizations (Assistant Secretary for Technology Policy/ONC, 2024). These limitations contribute to delays, errors, and administrative workload that can reduce the quality and timeliness of patient care (Centers for Medicare & Medicaid Services [CMS], 2020).

Delays in transferring records can cause incomplete referrals, postponed appointments, and gaps in care (Patel et al., 2018). Staff must also spend additional time managing requests, resending documents, and tracking referrals (ONC, 2020). These challenges are greater in rural and under-resourced areas where limited infrastructure and connectivity make data exchange more difficult (Assistant Secretary for Technology Policy/ONC, 2024). Missing information may also contribute to delayed diagnoses and unnecessary repeat testing (De Swart et al., 2023).

Current systems have improved but still lack compatibility across vendors, often requiring providers to use multiple platforms to access patient information (ONC, 2022; ONC, 2023). Fragmented systems can also create challenges in securely handling patient information and maintaining Health Insurance Portability and Accountability Act (HIPAA) compliance. A secure and efficient solution is therefore needed to improve data exchange without replacing existing electronic health record (EHR) systems.

PatientPath addresses these challenges through a secure, web-based platform for real-time medical record sharing and referral coordination. By connecting existing systems, PatientPath aims to improve interoperability, reduce delays, and increase the efficiency and quality of patient care.

2 Product Description

PatientPath is a secure, web-based platform designed to improve how healthcare organizations exchange patient medical records and manage referrals. Instead of relying on fax machines and other outdated methods, PatientPath allows healthcare providers to securely send and receive patient information digitally. The platform is designed to work alongside existing electronic health record (EHR) systems rather than replace them, making it easier for different healthcare organizations to share information. Providers, administrative staff, and patients can access features based on their assigned roles, while security measures protect sensitive patient information.

The main goal of PatientPath is to reduce delays, lost documents, and administrative work associated with traditional medical record transfers. The system supports real-time record sharing, referral tracking, patient consent management, and secure access to medical information. PatientPath also supports healthcare interoperability standards such as FHIR and HL7, allowing information from different EHR systems to be exchanged more easily. Additional security and reliability features, including encryption, role-based access control, multi-factor authentication, audit logging, and offline synchronization, help PatientPath provide a secure and

accessible solution for healthcare organizations, including those in rural or low-connectivity environments.

2.1 Key Product Features and Capabilities

PatientPath provides several features that improve the process of transferring medical records and coordinating patient referrals. **Secure record sharing** allows authorized healthcare organizations to digitally exchange medical information without relying on fax machines. Records are encrypted to protect sensitive patient information during storage and transmission.

The platform also provides **automated referral tracking**, allowing providers and administrative staff to monitor a referral from its creation through completion. Status updates and notifications help identify referrals that have not been addressed and reduce the need for staff to manually contact other organizations for updates. **Patient consent management** gives patients greater control over who is authorized to access their medical information.

PatientPath includes **multi-factor authentication and role-based access control** to prevent unauthorized access. Users receive permissions based on their role, such as patient, provider, specialist, or administrator. The system also maintains **audit logs** that record important actions involving patient information, providing accountability and supporting HIPAA-related security requirements.

Another important capability is **FHIR and HL7 integration**, which allows PatientPath to communicate with different EHR systems and convert information into a standardized format. PatientPath also provides **offline synchronization**, allowing supported information and actions

to remain available during temporary connectivity problems and synchronize when a connection is restored.

The combination of these features makes PatientPath different from systems that simply replace traditional faxing with digital faxing. PatientPath acts as a bridge between existing healthcare systems while combining record exchange, referral management, patient consent, security, and offline accessibility within one platform. This approach addresses the delays and communication problems caused by fragmented medical record exchange without requiring healthcare organizations to replace their existing EHR systems.

Table 1. PatientPath Features and Purposes

Features	Purpose
Secure Record Sharing	Allows authorized organizations to securely exchange medical records digitally.
Automated Referral Tracking	Tracks referrals from creation through completion.
Patient Consent Management	Allows patients to control access to their medical information.
MFA & Role-Based Access Control	Helps prevent unauthorized access based on assigned user roles.
Audit Logs	Records important actions involving patient information.
FHIR & HL7 Integration	Supports standardized exchange with different EHR systems.
Offline Synchronization	Supports temporary connectivity loss and synchronization when the connection returns.

2.2 Major Components (Hardware/Software)

PatientPath does not require healthcare organizations to purchase specialized hardware. Because it is a web-based system, users can access the platform through standard computers, tablets, or other supported devices with a modern web browser. The system is hosted using cloud infrastructure that provides the computing, database, document storage, and security resources required to operate the platform. Cloud-based storage and key management services are used to securely store medical records and protect encryption keys.

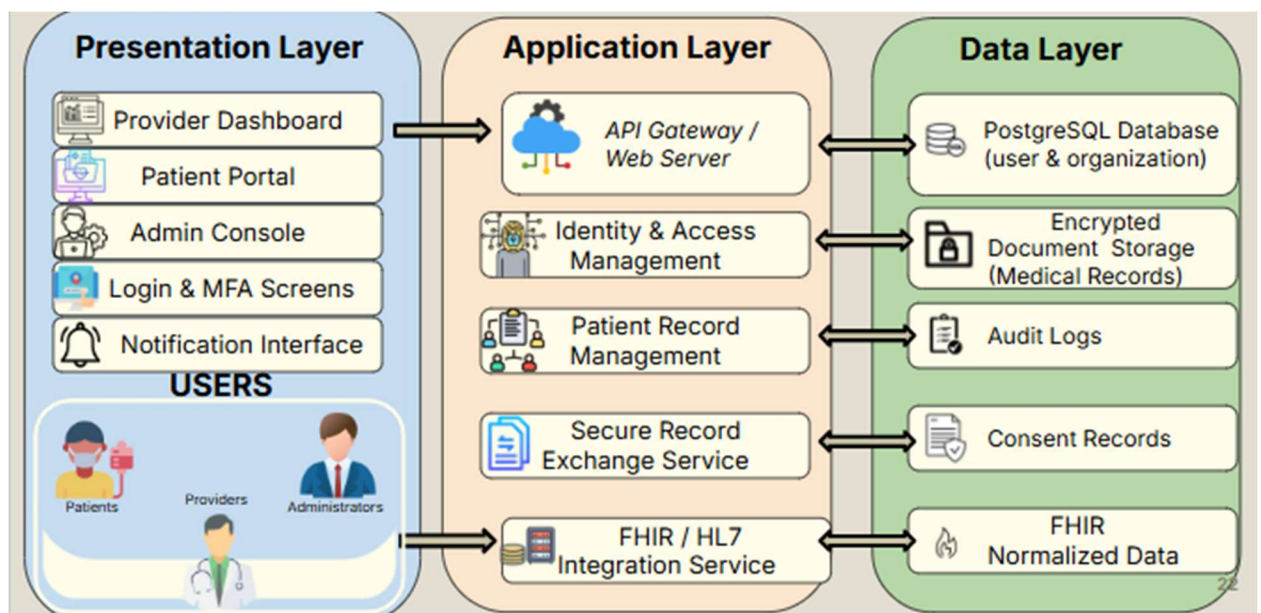


Figure 1. PatientPath Major Functional Components Diagram

As represented in the PatientPath Major Functional Component Diagram (MFCD), the software is organized into three primary layers: the **Presentation Layer**, **Application Layer**, and **Data Layer**. The Presentation Layer contains the interfaces used by patients, providers,

specialists, and administrators. These interfaces allow users to upload and access records, manage referrals, provide consent, and perform other actions based on their assigned roles.

The Application Layer contains the main logic of PatientPath. It manages secure record exchange, user authentication and authorization, patient records, referrals, and communication with existing EHR systems through FHIR and HL7. The Data Layer manages the information required by the platform, including user and organization data, medical records, consent information, and audit logs.

The PatientPath software will use **React and TypeScript** for the user interface and **Node.js with Express** for the server-side application. **PostgreSQL** will manage structured application data, while secure cloud storage will be used for medical documents. Additional technologies will support multi-factor authentication, encryption, real-time updates, offline synchronization, and FHIR/HL7 interoperability. Together, these components provide the software foundation needed for PatientPath to securely exchange medical information and coordinate referrals between healthcare organizations.

3 Identification of Case Study

PatientPath is being developed primarily for small-to-mid-sized healthcare organizations, including primary care practices, specialty clinics, and federally qualified health centers (FQHCs). These organizations may experience difficulties exchanging patient records and coordinating referrals with outside providers, especially when they rely on fax machines or disconnected healthcare systems. PatientPath is intended to provide these organizations with a secure and efficient way to exchange medical information without requiring them to replace their existing electronic health record (EHR) systems. The platform is also designed to support rural and under-resourced healthcare organizations that may experience limited technology resources or unreliable network connectivity.

The case study group for the PatientPath prototype will represent a small primary care clinic working with an outside specialty clinic. The group will include a primary care provider who creates referrals, a specialist who receives and reviews referrals, administrative staff who coordinate referrals and patient records, a system administrator who manages users and system activity, and a patient who manages consent and views the status of their information. This group represents the main users who would interact with PatientPath during a typical referral process. Feedback from these users will help the development team evaluate the usability and functionality of the prototype and identify areas that need improvement.

In the case study scenario, a patient visits their primary care provider and needs to be referred to an outside specialist. The provider uses PatientPath to create the referral and securely share the necessary patient records with the specialist. Administrative staff can monitor the referral and

make sure the required information has been sent. The specialist can access the referral and medical records based on their assigned permissions and update the referral status as the patient moves through the process. The patient can manage consent and view the status of the referral. Throughout the process, important system activities are recorded so administrators can review how patient information is accessed and shared. This scenario demonstrates several of PatientPath's main capabilities, including secure record sharing, referral tracking, role-based access control, patient consent, and audit logging.

In the future, PatientPath could be expanded for use by a wider range of healthcare organizations. Potential users include behavioral health providers, long-term care facilities, multi-specialty practices, regional health information exchanges (HIEs), and larger hospital systems. A full implementation could also expand PatientPath's interoperability with additional healthcare systems and allow more organizations to securely participate in the referral and record-sharing process. Expanding to these organizations would allow PatientPath to support medical record exchange and referral coordination across a broader healthcare network.

4 Glossary

- **EHR:** Electronic Health Record
- **HIPAA:** Health Insurance Portability and Accountability Act
- **FHIR:** Fast Healthcare Interoperability Resources (HL7 standard for health data exchange)
- **HL7:** Health Level Seven (International healthcare data standards organization)
- **RBAC:** Role-Based Access Control
- **MFA:** Multi-Factor Authentication
- **API:** Application Programming Interface
- **REST:** Representational State Transfer (architectural style for APIs)
- **HIE:** Health Information Exchange
- **PCP:** Primary Care Provider
- **FQHC:** Federally Qualified Health Center
- **CI/CD:** Continuous Integration / Continuous Deployment

5 References

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